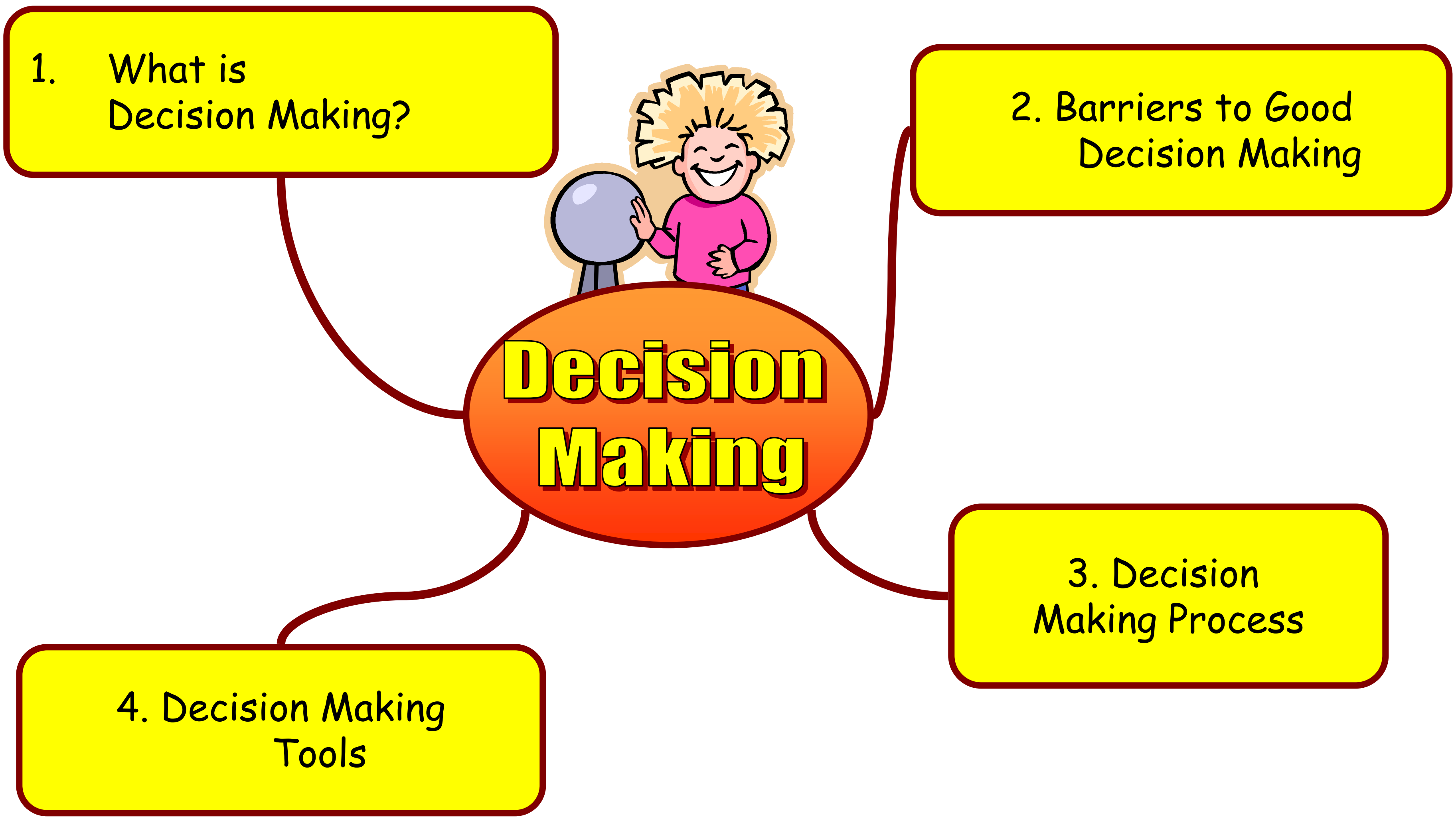
ENGINEERING ECONOMICS

What is Decision Making?

- **We all make decisions of varying importance every day, so the idea that decision making can be a rather sophisticated art is not valid**
  - **Studies have shown that most people are much poorer at decision making than they think!.**
- **An understanding of what decision making involves, together with a few effective techniques, will help produce better decisions**

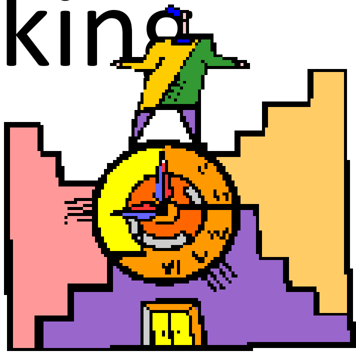
Decision Making



What is Decision Making?

- **Decision making** is the cognitive process leading to the selection of a course of action among alternatives.
- Every decision-making process produces a **final choice**. It can be an **action** or an **opinion**.
- It begins when we need to do something, but we do not know what. Therefore, decision making is a **reasoning process** which can be rational or irrational and can be based on explicit assumptions or tacit assumptions.

Barriers to Good Decision Making



- **Hasty** - Making quick decisions without having much thought.
- **Narrow** - Decision making is based on very limited information.
- **Scattered** - Our thoughts in making decisions are disconnected or disorganized.
- **Fuzzy** - Sometimes, the lack of clarity on important aspects of a decision causes us to overlook certain important considerations.

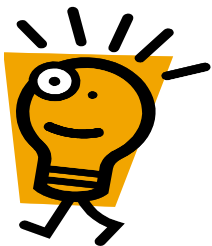
Decision Making Process

Steps	Tips
1. Identify a problem or opportunity.	❑ A lot of decision making goes wrong at the starting point. ❑ The more specific your definition of the decision is to made, the clearer will be your analysis and the likelihood of success.
2. Gather Information 3. Analyze Situation 4. Develop Options	❑ Successful decision makers explore all of the possible choices of the situation. ❑ In fact, many of the less obvious choices turn out to be the most effective ones. ❑In many cases, we may lack sufficient information to make an informed decision.
5. Evaluate Alternatives	❑ Analyze the advantages and disadvantages of each choice
6. Select the preferred alternatives	❑ Synthesize all what you learned in previous steps and make a conclusion that you believe to be your “best” choice.
7. Act on decision and monitor the results, making necessary adjustments.	❑ Once you have selected your best choice, you need to develop and implement a specific and concrete plan of action. ❑ As you begin taking the steps in your plan, you will discover that adjustments need to be made.

Decision Making Tools

Tools to facilitate better decisions:

- SWOT ANALYSIS
- SIX THINKING
- DECISION MATRIX
- FORCE FIELD ANALYSIS
- ISHIKAWA DIAGRAM
- MIND MAPPING
- SOCRATIC QUESTIONING



SWOT Analysis

- **SWOT** analysis is a great technique for identifying your **Strengths** and **Weaknesses** and study any **Opportunities** and **Threats** you face.
- It is also a powerful **strategic planning tool** used to evaluate a project or in a business venture or in any other situation of an organization or individual requiring a decision in pursuit of an objective.
- It involves monitoring the Management environment **internal** and **external** to the organization or individual.

SWOT Analysis – Strategic Use

- **Orienting SWOTs to An Objective** - If SWOT analysis does not start with defining a desired end state or objective, it runs the risk of being useless.
- If a clear objective has been identified, SWOT analysis can be used to help in the pursuit of that objective. In this case, SWOTs are:

Strengths Attributes of the organization that are helpful to achieving the objective.	**Weaknesses** Attributes of the organization that are harmful to achieving the objective.
Opportunities External conditions that are helpful to achieving the objective.	**Threats** External conditions that are harmful to achieving the objective.

SWOT Analysis – Creative Use

- ***Creative Use of SWOTs* –** If the objective seems attainable, the SWOTs are used as inputs to the creative generation of possible strategies, by asking (usually in groups) and answering each of the following four questions, many times:

Strengths How can we **U**se each Strength?	**Weaknesses** How can we **S**top each Weakness?
Opportunities How can we **E**xploit each Opportunity?	**Threats** How can we **D**efend against each Threat?

SWOT Analysis – Internal and External Factors

- The aim of any SWOT analysis is to identify the key ***internal*** and ***external*** factors that are important to achieving the objective. SWOT analysis groups key pieces of information into two main categories:

Internal Factors
➔ Organization

- Strengths
- Weaknesses

External Factors
➔ External Environment

- Opportunities
- Threats

SWOT Analysis – Errors to Be Avoided

- Conducting a SWOT analysis *before* defining and agreeing upon an objective (a desired end state). SWOTs should not exist in the abstract. They can exist only with reference to an objective.
- Opportunities external to the company are often confused with strengths internal to the company. They should be kept separate.
- SWOTs are sometimes confused with possible strategies. SWOTs are descriptions of conditions, while possible strategies define actions.

Strengths 1. What is our competitive advantage? 2. What resources do we have? 3. What products are performing well?	**Weaknesses** 1. Where can we improve? 2. What products are underperforming? 3. Where are we lacking resources?
Opportunities 1. What new technology can we use? 2. Can we expand our operations? 3. What new segments can we test?	**Threats** 1. What regulations are changing? 2. What are competitors doing? 3. How are consumer trends changing?